Commands cheat sheet

On the Jetson, in `~/GrappStar/jetson`. One program on the base port at a time. Full lists: `firmware/base/PROTOCOL.md`, `firmware/dart/API.md`, `jetson/README.md`. If a command here and those files disagree, those files win.

Stop: 1. `estop` (or STOP on the operator page). 2. Pull lead #2's XT60 (motors off). 3. Main battery. Cutting power does NOT release a drawn spring or a taut line. **Before walking up:** `estop`, lead #2 out, `cock` 0 and `H` closed, line slack, dart IDLE or its loop plug out. Never fire without the tether. Stand beside, never in line with, a towing line.

Jetson

Command	What it does
`python3 -m grappstar serve`	operator page at http://<jetson>:8080 (`--sim` for no hardware)
`python3 -m grappstar base ping` / `status` / `monitor`	talk to the base; `monitor` shows live telemetry (Ctrl+C stops)
`python3 -m grappstar base console`	live telemetry and events; type commands (JSON, `mode to=REEL`, `stop`); `quit`
`python3 -m grappstar base cmd '<json>'`	send one base command (below)
`python3 -m grappstar base mode LAUNCH` / `base aim 30` / `base fire`	shift, aim, fire (fire asks you to type FIRE)
`python3 -m grappstar dart status`	one dart status reading
`python3 -m grappstar dart key 2 --hold 1.0`	hold a dart key for 1 s (quote `'!'`, `'#'`, `'$'`)
`python3 -m grappstar dart auto` / `release` / `flightlog`	anchor, release, download the flight log
`python3 -m grappstar cameras`	list the cameras
`sudo shutdown now`	before unplugging the battery

Base commands (JSON, for `base cmd`)

Command	What it does
`{"cmd":"estop"}`	wiper off, drum to NEUTRAL when safe. Always safe to send
`{"cmd":"stop"}`	wiper off, drum stays
`{"cmd":"clear"}`	clear a fault once its cause is gone
`{"cmd":"mode","to":"LAUNCH"}`	drum to LAUNCH, NEUTRAL or REEL
`{"cmd":"aim","deg":30}`	rail to 30 degrees (only with the carriage home)
`python3 -m grappstar base fire` (Jetson CLI)	arm and fire in one go after you type FIRE. Tether on, area clear
`{"cmd":"decock"}`	unload a cocked carriage slowly (`"force":true` if SW_HOME is broken)
`{"cmd":"wrap"}`	lay 7 wraps, right after landing (from LAUNCH or NEUTRAL; parks at WRAPPED)
`{"cmd":"tidy"}`	take up the slack after the dart anchors
`{"cmd":"unwrap"}`	remove the wraps (after reeling in to about 1.5 m)
`{"cmd":"retrieve","duty":0.3}`	final approach on the take-up until the dart seats
`{"cmd":"reel","duty":0.5,"until":"seated"}`	wind in until the dart seats (`until`: stop, seated, tension)
`{"cmd":"jog","motor":"wiper","dir":1,"ms":300}`	short move; `motor` wiper or drum, `dir` 1 or -1, up to 1500 ms
`{"cmd":"set","key":"limit_n","val":40}`	set a stored value (tow limit; 40 N or less for the first field tows)
`{"cmd":"get","key":"limit_n"}`	read a stored value

Dart keys (WiFi KEDGE, password grappstar, http://192.168.4.1, or Serial 115200)

Key	Motor
1 / !	deploy pods down / up
2 / @	kick out / back
3 / #	drills in / out
4 / $	foot in / out
space	STOP everything
a / A	auto anchor now / arm auto-anchor on landing
i / b / z	current / attitude / I2C scan